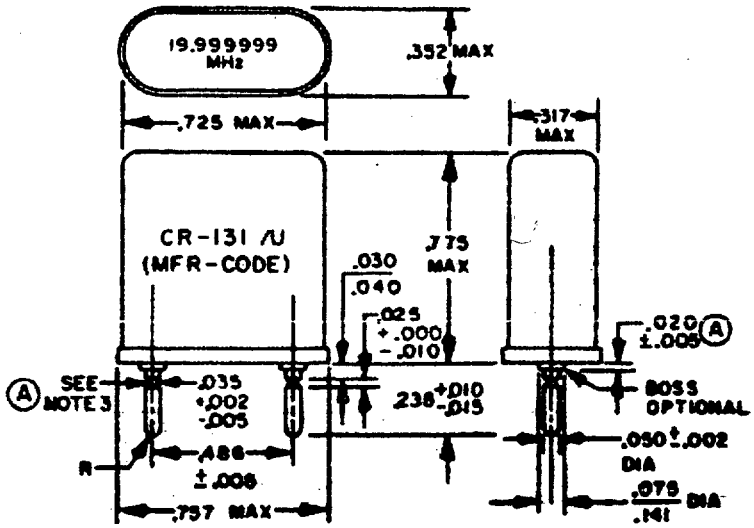
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-131/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue Specification MIL-C-3098.

Pertinent characteristics: 0.8 to 20 MHz; fundamental; controlled; antiresonance.



INCHES	MM	INCHES	MM
.002	.05	.050	1.27
.005	.13	.075	1.91
.008	.20	.141	3.58
.010	.25	.238	6.05
.015	.38	.317	8.05
.020	.51	.352	8.94
.025	.64	.486	12.34
.030	.76	.725	18.42
.035	.89	.757	19.23
.040	1.02	.775	19.69

NOTES:
1. Dimensions are in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
(A) 3. The pin undercut may be omitted.

(A) denotes changes

FSC 5955

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Ⓐ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-6/U (see MIL-H-10056/2)
Dimensions, marking, and configuration	See figure
Frequency range	0.8 to 20.0 MHz, inclusive
Mode of oscillation	Fundamental
Antiresonance, load capacitance	30.0 ±0.5 pF
Test set and calibration values	See table II
Shock (specified pulse):	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Rated drive level	See table II
Capacitance, shunt	7 pF, maximum
Frequency and equivalent resistance	See table III
Reference temperature	75° ±1°C
Frequency tolerance	±20 ppm
Operatable temperature range	-55° to +70° and 80° to 90°C, inclusive
Operating	70° to 80°C, inclusive
Frequency stability	±5 ppm
Temperature run:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	Below 2 MHz - ±15 percent 2 MHz and above - ±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±15 percent

1/ Method 201 of MIL-STD-202

TABLE II. Test set, calibration values, and rated drive level.

Government reference standard test set	Band of specified frequencies, inclusive	Calibration values			Rated drive level
		Resistance	Crystal current	Resistor voltage drop	
	MHz	Ohms	mA	Volt	mV
TS-330/TSM	0.8 to 1.5	0	1/	---	5.0 + 1.0 2/
	1.5+ to 3.4	50	10	---	5.0 + 1.0
	3.4+ to 5.1	22	15	---	5.0 + 1.0
	5.1+ to 10	13	20	---	5.0 + 1.0
	10+ to 15	11	15	---	2.5 + 0.5
TS-683/TSM	15+ to 20	10	---	0.16	2.5 0.5

1/ Set crystal current control at extreme counterclock (minimum) position.
2/ The violation of $P=I^2R$ is intentional; at the higher resistance of the crystal unit, the empirical power dissipation will be as rated.

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TABLE III. Resistance, equivalent.

Frequency range inclusive			Maximum resistance
MHz			Ohms
0.8	to	0.85	630
0.85+	to	0.9	600
0.9+	to	1	580
1+	to	1.12	540
1.12+	to	1.25	490
1.25+	to	1.37	450
1.37+	to	1.5	410
1.5+	to	1.62	380
1.62+	to	1.75	330
1.75+	to	1.87	300
1.87+	to	2	290
2+	to	2.12	270
2.12+	to	2.25	250
2.25+	to	2.6	200
2.6+	to	3	150
3+	to	3.4	110
3.4+	to	3.75	90
3.75+	to	4	75
4+	to	5	60
5+	to	7	35
7+	to	10	24
10+	to	15	22
15+	to	20	20

NOTES:

Ⓐ 1. The socket designed to mate with crystal holder HC-6/U is Part No. MI2883/04-01 (see MIL-S-12883/4).
For Navy use, other sockets listed in MIL-STD-242 may be used.

2. NATO preferred type NXF7.

3. This crystal unit is identical to the CR-27A/U, except for a load capacitance rating of 30 pF ±0.5 pF. This crystal unit is not interchangeable with the CR-27A/U.

Custodian:
Army - EL

Review activities:
Army - MI, MU
DSA - ES

Preparing activity:
Army - EL

Agent:
DSA - ES

(Project 5955-A172-4)

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